

MODEL TEST PAPER 4
FOUNDATION COURSE
PAPER – 3: QUANTITATIVE APTITUDE

Time: 2 Hours

Marks: 100

1. P, Q and R three cities. The ratio of average temperature between P and Q is 11: 12 and that between P and R is 9:8. The ratio between the average temperature Q and R
 - (a) 22 : 27
 - (b) 27 : 22
 - (c) 32 : 33
 - (d) none of these
2. The third proportional between (a^2-b^2) and $(a+b)^2$ is :
 - (a) $\frac{a+b}{a-b}$
 - (b) $\frac{a-b}{a+b}$
 - (c) $\frac{(a-b)^2}{a+b}$
 - (d) $\frac{(a+b)^3}{a-b}$
3. If 8th term of an AP is 15, the Sum of the 15 its term is
 - (a) 15
 - (b) 0
 - (c) 225
 - (d) 225/2
4. For what values of x, the number $-\frac{2}{7}, x, -\frac{7}{2}$ are in G.P.?
 - (a) ± 1
 - (b) ± 3
 - (c) ± 2
 - (d) none of these
5. For what value of x; the sequence x+1, 3x, 4x+2 are in AP?
 - (a) 3
 - (b) 2

- (c) 4
(d) 5
6. If $a^{1/x} = b^{1/y} = c^{1/z}$ and a, b, c are in GP then x, y, z are in
(a) AP
(b) GP
(c) HP
(d) AGP
7. The derivative of $e^x \log x$
(a) $\frac{e^x}{x}(1 + x \log x)$
(b) $\frac{e^x}{x}(1 + \log x)$
(c) $(1 + \log x)$
8. If $y = \sqrt{\frac{1-x}{1+x}}$ then $(1-x^2) \frac{dy}{dx} =$
(a) y
(b) $-x$
(c) $-y$
(d) 0
9. Find the gradient of the curve $y = 3x^2 - 6x + 4$ at the point $(1, 2)$
(a) 1
(b) -1
(c) 0
(d) 2
10. The equation of the curve in the form $y = f(x)$ if the curve passes through the point $(1, 0)$ and $f'(x) = 2x - 1$ is
(a) $y = x^2 - x$
(b) $x = y^2 - y$
(c) $y = x^2$
(d) none of these
11. $\int \frac{1}{x \log x} dx = ?$
(a) $\log|x| + c$

- (b) $\log |\log x| + c$
 (c) $(\log x)^2 + c$
 (d) none of these
12. $\int_1^2 \frac{2x}{1+x^2} dx$ is equal to
 (a) $\log_e(5/2)$
 (b) $\log_e 5 - \log_e 2 + k$
 (c) $\log_e(2/5)$
 (d) none of these
13. Find $f \circ g$ for the functions $f(x) = x^8$, $g(x) = 2x^2 + 1$
 (a) $x^8 (2x^2 + 1)$
 (b) x^8
 (c) $2x^2 + 1$
 (d) $(2x^2 + 1)^8$
14. The number of proper subsets of the set $\{3, 4, 5, 6, 7\}$ is
 (a) 32
 (b) 31
 (c) 30
 (d) 25
15. On the sets of lines in a plane the Relation "is perpendicular to" is
 (a) Reflexive
 (b) Symmetric
 (c) Transitive
 (d) none of these
16. In how many ways 3 prizes out of 5 can be distributed amongst 3 brothers equally
 (a) 10
 (b) 45
 (c) 60
 (d) 120
17. There 12 questions to be answered to be Yes or No. How Many ways this can be answered -
 (a) 1021

- (b) 2048
 - (c) 4096
 - (d) None of the above
18. ${}^{15}C_{3r} = {}^{15}C_{r+3}$, then r is equal to
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 5
19. A polygon has 44 diagonals then the number of sides are
- (a) 6
 - (b) 7
 - (c) 8
 - (d) 11
20. The number of ways of painting the six faces of a cube with six different given colours is
- (a) 1
 - (b) 720
 - (c) 30
 - (d) 15
21. How many Six-digit telephone numbers can be formed by using 10 distinct digits
- (a) 10^8
 - (b) 6^{10}
 - (c) ${}^{10}C_9$
 - (d) ${}^{10}P_6$
22. ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots =$
- (a) $2^n - 1$
 - (b) 2^n
 - (c) 2^{n+1}
 - (d) none of these
23. The value of $\log_{0.1} 0.001 =$
- (a) 3
 - (b) 2

- (c) 4
(d) $\frac{1}{3}$
24. if $\log_4 x = -\frac{3}{2}$. Then x is
(a) $\frac{1}{8}$
(b) $\frac{1}{4}$
(c) $\frac{1}{2}$
(d) $\frac{1}{3}$
25. A number consists of two digits. The digits in tens place is 3 times the digit in the unit's place. If 54 is subtracted from the digits are reversed. The number is
(a) 39
(b) 92
(c) 93
(d) 94
26. The equation $x^2 - (P+4)x + 2P+5 = 0$ has equal roots
The value of p is
(a) 2
(b) -2
(c) ± 2
(d) 3
- 27.

x	5	6	7	8
y	11	13	15	17

- In the above table corresponding values of two variable x and y have been given. Which of the following equations establishes the relationship between the two variables?
(a) $y = 3x+2$
(b) $y = 2x-1$
(c) $y = 2x+1$
(d) $y = 3x+1$
28. A manufacturer produces two items A and B. He has ₹ 10,000 to invest and a space to store 100 its ms. A table costs him ₹ 400 and a chair ₹ 100. Express this in the form of linear inequalities.
(a) $x + y \leq 100$, $4x + y \leq 100$, $x \geq 0$, $y \geq 0$

- (b) $x + y \leq 1000$, $2x + 5y < 1000$, $x \geq 0$, $y \geq 0$
- (c) $x + y > 100$, $4x + y \geq 100$, $x \geq 0$, $y \geq 0$
- (d) none of these
29. The difference between compound and simple interest at 5% per annum for 4 years on ₹ 20,000 is -
- (a) 250
- (b) 277
- (c) 300
- (d) 310
30. In how many years will a sum of money double at 5% p.a compounded interest?
- (a) 15 years 3 months
- (b) 14 years 2 months
- (c) 14 years 3 months
- (d) 15 years 3 months
31. A machine worth ₹ 4,90,740 is depreciated at 15% of its opening value each year. When would its value reduce by 90%?
- (a) 11 years 6 months
- (b) 11 years 7 months
- (c) 11 years 8 months
- (d) 14 years 2 months approximately
32. Assuming, that discount rate is 7% per annum, how much would you pay to receive ₹ 50, growing at 5%, annually, forever.
- (a) 2500
- (b) 3000
- (c) 3500
- (d) 4000
33. Future value of Ordinary Annuity
- (a) $A(n, i) = A \left[\frac{(1+i)^n - 1}{i} \right]$
- (b) $A(n, i) = A \left[\frac{(1+i)^n + 1}{i} \right]$

$$(c) \quad A(n, i) = A \left[\frac{1 - (1+i)^n}{i} \right]$$

$$(d) \quad A(n, i) = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

34. Nominal rate of Interest 9.9% p.a. If Interest is compounded monthly. What will be the effective rate of Interest? (Given $\left(\frac{4033}{4000}\right)^{12} = 1.1036$)
- (a) 10.36 %
 - (b) 9.36%
 - (c) 11.36%
 - (d) 9.9 %
35. A machine worth of ₹ 4,90,740 is depreciated at 15% on its opening value each year. When its value reduces to ₹ 2,00,000
- (a) 4 years 6 months
 - (b) 4 years 7 months
 - (c) 4 years 5 months
 - (d) 5 years 7 months approximately
36. A sinking fund is created redeeming debentures worth ₹ 5,00,000 at the end of 25 years. How much provision need to be made out of profits each year provided sinking fund investments can earn at 4 % per annum
- (a) 12,006
 - (b) 12,040
 - (c) 12039
 - (d) 12035
37. Nominal Rate of Return =
- (a) Real Rate of Return – Inflation
 - (b) Real Rate of Return + Inflation
 - (c) Real Rate of Return / Inflation
 - (d) Real Rate of Return × Inflation
38. Net Present value ≥ 0 , then
- (a) Accept the Proposal
 - (b) Reject the proposal
 - (c) Not Feasible

- (d) None of the above
39. A sum of Money doubles itself at compound interest in 10 years. In how many years will it become eight times
- (a) 10
 - (b) 30
 - (c) 40
 - (d) 35
40. The time in which a sum of money will be doubled at 6% compound interest compounded annually approximately.
- (a) 10 years
 - (b) 12 years
 - (c) 13 years
 - (d) 14 years
41. 18, 24, 21, 27, ?, 30, 27
- (a) 33
 - (b) 30
 - (c) 24
 - (d) 21
42. 5, 7, 11, ?, 35, 67
- (a) 23
 - (b) 28
 - (c) 30
 - (d) 19
43. If GARDEN is coded as 325764 and WATER as 92165, how can we code the word WARDEN in the same way?
- (a) 925764
 - (b) 295764
 - (c) 952764
 - (d) 957264
44. If F = 6, MAT=34, then how much is CAR?
- (a) 21
 - (b) 22
 - (c) 25

- (d) 28
45. Find next term of the series, 4, 9, 16, 25, 36, 49, ?
- (a) 1
(b) 9
(c) 20
(d) 64
46. Find odd man out of the series 16, 25, 36, 72, 144, 196, 225
- (a) 36
(b) 72
(c) 196
(d) 225
47. Raju starts from point A and walks 1 km towards south, turns left and walks 1 km. Then he turns left again and walks 1 km. now he is facing?
- (a) East
(b) West
(c) North
(d) South-West
48. Roopa starts from a point and walks 15 metre towards west, turns left and walks 12 metre, turns right again and walks. What is the direction she is now facing?
- (a) South
(b) West
(c) East
(d) North
49. A car travelling from south to north covers a distance of 8 kms, then turns right and runs another 9 kms and again turns to the right and was stopped. Which direction does it face now?
- (a) South
(b) North
(c) West
(d) East
50. There are five houses P, Q, R, S and T. P is right of Q and T is left of R and right of P. Q is right of S. Which house is in the middle?
- (a) P

- (b) Q
 - (c) T
 - (d) R
51. Six friends A, B, C, D, E and F are sitting in a row facing towards North, C is sitting between A and E, D is not at the end, B is sitting at immediate right of E, F is not at the right end, but D is sitting at 3rd left of E. Which of the following is sitting to the left of D?
- (a) A
 - (b) F
 - (c) E
 - (d) C
52. Six girls are standing in such a way that they form a circle, facing the centre. Subbu is to the left of Pappu, Revathi is between Subbu and Nisha, Aruna is between Pappu and Keerthana. Who is to the right of Nisha?
- (a) Ravathi
 - (b) Aruna
 - (c) Subbu
 - (d) Keerthana
53. A is B's brother. C is D's father. E is B's mother. A and D are brothers. How is E related to C?
- (a) Sister
 - (b) Sister-in-law
 - (c) Niece
 - (d) Wife
54. A is B's brother, C is A's mother, D is C's father, E is B's son, How is B related to D?
- (a) Son
 - (b) Granddaughter
 - (c) Grandfather
 - (d) Great Grandfather
55. A is the mother of D and sister of B. B has a daughter C who is married to F. G is the husband of A. How is G related to D?
- (a) Uncle
 - (b) Husband
 - (c) Son

- (d) Father
56. P and Q are brothers. R and S are sister. P's son is S's brother. How is Q related to R?
- (a) Uncle
(b) Brother
(c) Father
(d) Grandfather
57. Pointing out to a photograph, a man tells his friend, "She is the daughter of the only son of my father's wife." How is the girl in the photograph related to the man?
- (a) Daughter
(b) Mother
(c) Cousin
(d) Sister
58. A party consists of grandmother, father, mother, four sons and their wives and one son and two daughters to each of the sons. How many females are there is all?
- (a) 13
(b) 16
(c) 18
(d) 24
59. Shyam goes 5 km in the North from his school. Now, turning to the left, he goes to 10 km and again turns to left and goes to 5 km. How far he is from his school and in which direction?
- (a) 10 km, South from school
(b) 10 km, North from school
(c) 10 km, West from school
(d) 10 km, East from school
60. Rasik walked 20 m towards north. Then he turned right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Finally, he turns left and walks 15 m. In which direction and how many metres is he from the starting position?
- (a) 15 m West
(b) 30 m East
(c) 30 m West

- (d) 45 m East
61. The _____ is satisfied when $P_{ab} \times P_{bc} \times P_{ca} = 1$
- Time reversal test
 - Factor reversal test
 - Circular Test
 - none of these
62. The index number of prices at a place in 2008 is 355 with 2003 as base. This means -
- There has been on the average a 255% increase in prices.
 - There has been on the average a 355% increase in price.
 - There has been on the average a 250% increase in price.
 - None of these.
63. The number of tests of Adequacy
- 2
 - 3
 - 4
 - 5
64. If two events A and B are independent, the probability that both will occur is given by
- $P(A) \times P(B)$
 - $P(A) + P(B)$
 - $P(A) + P(B) - P(A \cup B)$
 - $P(A) + P(B) - P(A \cap B)$
65. If $p:q$ is the odds in favor of an event, then the probability of that event is -
- p/q
 - $\frac{q}{p+q}$
 - $\frac{p}{p+q}$
 - none of these
66. If $P(A) = 4/9$; then the odd against the event 'A' is
- 4:9
 - 4:5
 - 5:4

- (d) 4:14
67. If two letters are taken at random from the word HOME, what is the Probability that none of the letters would be vowels?
- (a) $1/6$
(b) $1/2$
(c) $1/3$
(d) $1/4$
68. Equations of two lines of regression are $4x+3y+7 = 0$ and $3x+ 4y + 8 = 0$, the mean of x and y are
- (a) $5/7$ and $6/7$
(b) $-4/7$ and $-11/7$
(c) 2 and 4
(d) None of these
69. Correlation Co-efficient is _____ of the units of measurements
- (a) Independent
(b) Dependent
(c) Both
(d) none of these
70. If for two variable x and y, the covariance, variance of x and variance of y are 40, 16 and 256 respectively, what is the value of the correlation coefficient?
- (a) 0.01
(b) 0.625
(c) 0.4
(d) 0.5
71. Statistics is concerned with
- (a) Qualitative information
(b) Quantitative information
(c) (a) or (b)
(d) Both (a) and (b).
72. The standard deviation of 25, 32, 43, 53, 62, 59, 48, 31, 24, 33 is
- (a) 13.23
(b) 12.33
(c) 11.33

- (d) none of these
73. The quartile deviation of a normal distribution with mean 10 and standard deviation 4 is
- (a) 0.675.
 - (b) 67.50.
 - (c) 2.70
 - (d) 3.20.
74. If the range of x is 2, what would be the range of $-3x + 50$?
- (a) 2
 - (b) 6
 - (c) -6
 - (d) 44
75. If the quartile deviation of a normal curve is 4.05, then its mean deviation is
- (a) 5.26
 - (b) 6.24
 - (c) 4.24
 - (d) 4.80
76. The mean of first 3 terms is 14 and the mean of next 2 terms is 18. The mean of 5 numbers is -
- (a) 14.5
 - (b) 15
 - (c) 14
 - (d) 15.6
77. The Standard deviation is independent of change of
- (a) Origin
 - (b) Scale
 - (c) Both
 - (d) none
78. If two variables are uncorrelated then regression lines are
- (a) Parallel
 - (b) Perpendicular
 - (c) Coincident
 - (d) Inclined at 45°

79. When 'p' = 0.5, the
- Asymmetrical.
 - Symmetrical.
 - Both of above.
 - None of above
80. In a normal distribution skewness is ____
- 0
 - >3
 - <3
 - <1
81. If mean and standard deviation of a binomial distribution is 10 and 2 respectively; q will be ____
- 1
 - 0.8
 - 0.6
 - 0.4
82. Which one is not a condition of Poisson model
- the probability of having failures in a small time interval is constant
 - the probability of having success more than one in a small time interval is very small
 - the probability of having success in this time interval is independent of time 't' as well as earlier success
 - the probability of having success in a small time interval (t, t+td) is Kt for a positive constant k.
83. In _____ distribution, mean = variance.
- Normal
 - Binomial
 - Poisson
 - none of these
84. The points of inflexion of the normal curve $f(t) = \frac{1}{4\sqrt{2\pi}} e^{-\frac{(t-10)^2}{32}}$ are
- 6, 14
 - 5, 15
 - 4, 16

- (d) none of these
85. The total area of the normal curve is the
- one
 - 50 percent
 - 0.50
 - any value between 0 and 1
86. 'Stub' of a table is the _____ part of the table describing the _____.
- Left, Columns
 - Right, Columns
 - Right, Rows
 - Left, Rows
87. The pair of averages whose value can be determined graphically.
- Mean and Median
 - Mode and Mean
 - Mode and Median
 - None of these
88. Find the Expected value of the following distribution
- | | | | | | |
|------|------|-----|-----|------|------|
| x | -20 | -10 | 30 | 75 | 80 |
| P(x) | 3/20 | 1/5 | 1/2 | 1/10 | 1/20 |
- 20.5
 - 21.5
 - 22.5
 - 24.5
89. The tests of shifting bases are called ____
- Unit test
 - Time reversal test
 - Circular test
 - None of these
90. Purchasing power of money is stated as _____ price index?
- Equal to
 - Unequal to
 - Reciprocal of

- (d) None of these
91. If $\sum P_0Q_0 = 1360$, $\sum P_nQ_0 = 1900$, then the Laspyres Index number is
- 71
 - 139.70
 - 175
 - 180
92. The difference between the upper and lower limit of a class is called
- Class Interval
 - Mid Value
 - Class Boundary
 - Frequency
93. A man travels from Delhi to Agra at an average speed of 30km per hour and back at an average speed of 60 km per hour. What's the average Speed.
- 48 Km/ hr
 - 40 km/hr
 - 45 km/hr
 - 35 km/hr
94. If the mean of frequency distribution is 100 and coefficient of variation is 45% then standard deviation is.
- 45
 - 0.45
 - 4.5
 - 450
95. if the mean and SD of X are a and b respectively, then the S.D of $\frac{x-a}{b}$ is
- a/b
 - 1
 - 1
 - ab
96. If one regression coefficient is greater than one, then other will be:
- More than one
 - Equal to one
 - Less than one

- (d) Equal to minus one
97. The maximum value of correlation coefficient is
- (a) 0
 - (b) 1
 - (c) -1
 - (d) none of these
98. What is exclusive Series
- (a) In which both upper and lower limit are not included in class frequency
 - (b) In which lower limit is not included class frequency
 - (c) In which upper limit is not included in class frequency
 - (d) None of the above
99. If the arithmetic mean between two numbers is 64 and the Geometric Mean between them is 16. The Harmonic mean between them is ____
- (a) 64
 - (b) 4
 - (c) 16
 - (d) 40
100. When the mean is 3.57 and mode is 2.13, then the value of median is ____
- (a) 3.09
 - (b) 5.01
 - (c) 5.01
 - (d) none of these.